

Accountability in Algorithmic Systems: From Principles to Practice

Darcia Wilkinson
dwilkinson@microsoft.com
Microsoft Research
USA

Deborah Raji
rajiin@berkeley.edu
University of California, Berkeley
USA

Angelika Strohmayer
angelika.strohmayer@northumbria.ac.uk
Northumbria University
UK

Kate Crawford
kate@microsoft.com
Microsoft Research
USA

Bogdana Rakova
bogdana@mozillafoundation.org
Mozilla Foundation
USA

Ethan Zuckerman
ezuckerman@umass.edu
University of Massachusetts at
Amherst
USA

Hanna Wallach
wallach@microsoft.com
Microsoft Research
USA

Ranjit Singh
ranjit@datasociety.net
Data and Society
USA

ABSTRACT

Growing concerns over the societal implications of artificial intelligence has motivated an interdisciplinary push towards mechanisms and tools that hold algorithmic systems accountable. Although there have been considerable strides around defining what it means to hold AI systems accountable, operationalizing those principles have created a barrage of challenges. Researchers, practitioners, and regulators have all raised concerns about the completeness of accountability methods and observed spikes in anxiousness about the potential risk of these tools being manipulated as rubber stamps of approval while harms continue to slip through the cracks. This interactive panel gathers researchers and practitioners with expertise in HCI, Responsible AI, Machine Learning, and Public Policy to critically discuss issues regarding accountability in algorithmic systems to reflect on potential opportunities for re-imagining scalable directions for accountability within these systems.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Security and privacy** → **Human and societal aspects of security and privacy**.

KEYWORDS

Accountability, algorithmic systems, artificial intelligence, responsible AI, policy

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1 INTRODUCTION

Algorithmic tools and systems have revolutionized the global economy while simultaneously introducing adverse impacts to society at large [1]. People have the option to have tools embedded with the latest advances in Artificial Intelligence (AI) at their finger tips while observing elements of AI being incorporated and enmeshed with almost every element of their lives. This entanglement has made it particularly challenging to escape the adverse impacts of these technologies. With the goal of supporting the development and deployment of safer technologies multiple actors - regulators, academics, private companies, civic organizations, journalists, human rights advocates - have pushed efforts to keep systems accountable through practices that better identify, assess the extent of, and minimize harms from algorithmic systems. This panel aims to *deeply engage in conversations that explore the strengths and weaknesses of approaches aimed at introducing accountability in algorithmic systems*.

Efforts to support this trajectory towards accountability have resulted in the development of principle-based tools and harm minimization practices such as algorithmic auditing, impact assessments, and automated risk minimization software [5, 10, 11]. However, scholars have argued that a fundamental step towards adoption and development of accountability tools involves the creation of guiding frameworks to inform practices. Although both of these general directions have been beneficial, there are significant gaps that threaten their effectiveness. The scope for what should be included as core principles is vast and vague. The Global Inventory of AI Ethics Guideline reports that there are over 160 guiding frameworks in existence [19] and this is in addition to emerging guiding frameworks and standards being implemented by private

companies¹. Moreover, recent work has pointed to inconsistencies with how harm minimization practices (such as audits) are conducted in ways that contribute to disparate levels of completeness in the evaluation of systems. This in turn makes it difficult to verify how well systems are audited [5, 16]. Challenges with poor definitions, a limited agreement and understanding of practices, and a lack of standardization have raised concerns about the pitfalls of focusing on *what* should be included (i.e. ethical principles) instead of *how* it should be done (i.e. putting those principles into practice) [14]. As such, this panel *will explore options that allow us to leverage promising methodologies with actionable pathways towards accountability*.

It is difficult to ignore the increasing reports of harm that highlight ethical and practical challenges as a result of AI systems being inherently prone to errors and biases. The scope of issues range from the individual, collective, or societal level [18]. On the individual level, a person may be on the receiving end of harm due to embedded biases in the system. For example, the reliance on facial recognition as evidence in the pursuit of criminal cases led to the arrest of a New York resident for stealing a pair of socks from TJ Maxx although the individual was signed in at a hospital for the birth of his child at the time the crime was committed [9]. Collective harm includes wide-spread risks to entire groups of people with disparate system performance based on identity [13] due to challenges such as reinforced stereotypes [3, 15] or limited representation in training data [4]. Examples of societal harm include but are not limited to risks of inequitable distribution of resources, threats to the stability of democracy, and widening existing social inequalities [2, 6, 8, 12]. Consequently, in an effort to respond to the wide and expanding range of issues on our societies, policymakers have begun pushing for more incorporation of harm minimization practices (such as audits) in an effort to push companies to implement methods that would evaluate systems. However, different approaches to governance, varying legislative priorities, the extent of enforcement, cultural differences, and concerns of insufficient protections are all heatedly debated [17]. Acknowledging the socio-technical nature of how harm occurs is important in steps towards the design of solutions that are human-centered. Lastly, this view informs another goal of the panel which is *to explore a wider scope to understand how harms from algorithmic systems persist for different populations and affect policy*.

In this panel, we provide a space to drive discussions and debates that challenge current boundaries and refine new perspectives. The event gathers interdisciplinary perspectives from five panelists with experiences across HCI, responsible AI, and policy. Panelists have studied varying aspects of accountability in multiple contexts including practically evaluating accountability tools such as audits, exploring policy implications with emerging regulations, and investigating how harms manifest in non-Western or sensitive contexts. With the wealth of knowledge from the panelists and active engagement from audience members, this panel seeks to encourage wider participation within the SIGCHI community to contribute towards safer and more accountable algorithmic systems.

¹Examples include but are not limited to: Microsoft's Responsible AI Standard (link: <https://www.microsoft.com/en-us/ai/responsible-ai>), and Google's AI Principles (link: <https://ai.google/principles/>)

2 PANEL OVERVIEW

The core questions that will guide the dialogue would include:

- What are the most pressing challenges hindering accountability in algorithmic systems?
- In what ways could we re-imagine methodologies and practices for incorporating accountability throughout the development cycle of algorithmic systems?
- What is missing from current or past work on accountability?
- How might we conceive technological futures that acknowledge diverging intercultural needs (albeit organizational culture or socio-cultural needs)?

3 PANEL COMPOSITION

3.1 Panel Organizers/Moderators

The organizers of this event have centered their careers on Fairness, Accountability, Transparency, and Ethics in AI. Combined, they have published across multiple SIGCHI venues geared at the intersection of HCI and AI. Below we offer a brief insight into the backgrounds of the organizers. For convenience, organizers are listed in alphabetical order:

Kate Crawford Crawford is a Senior Principal Researcher at Microsoft Research, where she co-founded the FATE Group, and a Research Professor at USC Annenberg, where she leads the Knowing Machines Project, an international research group that studies the foundations of machine learning. Crawford is the inaugural Visiting Chair for AI and Justice at the École Normale Supérieure in Paris, and is an Honorary Professor at the University of Sydney. Her latest book *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence* (Yale University Press, 2021) won the Sally Hacker Prize from the Society for the History of Technology, as the book of the year award from the Association for Information Science and Technology. Both the Financial Times and New Scientist described *Atlas of AI* as one of the best books of 2021. Crawford received the Ayrton Prize from the British Society for the History of Science, and her project *Anatomy of an AI System* with Vladan Joler won the Beazley Design of the Year Award, the S+T+ARTS Prize Honorary mention by the European Commission, and is in the permanent collection of the V&A in London and MoMA in New York (where it is currently on show from 2022-2024).

Hanna Wallach is a partner research manager at Microsoft Research New York City. Her research focuses on issues of fairness, accountability, transparency, and ethics as they relate to AI and machine learning. She collaborates with researchers from machine learning, natural language processing, human-computer interaction, and science and technology studies, as well as lawyers and policy makers; her research integrates both qualitative and quantitative perspectives. Previously, she developed machine learning and natural language processing methods for analyzing the structure, content, and dynamics of social processes. She collaborated with political scientists, sociologists, journalists, and others to understand how organizations function by analyzing publicly available interaction data, including email networks, document collections, press releases, meeting transcripts, and news articles.

Daricia Wilkinson is a Post-Doctoral Researcher working in the FATE (Fairness, Accountability, Transparency, and Ethics in AI)

group at the Microsoft Research. She is passionate about finding ways to design safety tools to make the internet a safer place for all people. As such, her work unites strings from Human-Computer Interaction and AI to highlight opportunities for fairness, transparency, and justice online. Prior to joining MSR, Daricia received her Ph.D. in Human-Centered Computing at Clemson University under the advisement of Dr. Bart Knijnenburg. Originally from the Caribbean, she conducted work on the ground with a local non-profit in the Caribbean to complete her dissertation on alternative pathways for the design of justice-oriented safety countermeasures particularly for people in non-Western contexts.

3.2 Confirmed Panelists

All of the panelists have extensive experience studying accountability within AI systems. Panelists are top researchers with expertise that would allow a variety of perspectives about design, development and policy considerations. Our panelists also offer diverse representations in background and lived experiences as they all have unique intersectional identities coming from different countries while having deeply studied the impact of intelligent systems in different geographical regions including populations in the Global South. Below we offer brief biographical sketches for each panelist appearing in alphabetical order.

Deborah Raji is a Mozilla fellow and C.S. Ph.D. student at the University of California, Berkeley, who is interested in questions on algorithmic auditing and evaluation. She has worked closely with the Algorithmic Justice League to highlight bias in deployed AI products, worked with Google's Ethical AI Team, and been a research fellow at the AI Now Institute and Partnership on AI working on various projects to operationalize ethical considerations in machine learning engineering practice. She was recently named to *Forbes* 30 Under 30 and *MIT Tech Review* 35 Under 35 Innovators.

Bogdana Rakova is a Senior Trustworthy AI Fellow at Mozilla Foundation. She is an engineer, researcher, inventor, and organizer. Her work investigates the complex issues at the intersection of people, technology, trust, transparency, accountability, and social and environmental justice. Previously, she was a research manager at Accenture's Responsible AI team, a mentor at the Assembly Ethics and Governance of AI program led by Harvard's Berkman Klein Center for Internet and Society, and has also held positions with Partnership on AI, Samsung Research America, and others. She was also part of the core team at the IEEE P7010 working group developing a recommended practice for assessing the impact of AI on human well-being and consequently joined the board at the Happiness Alliance nonprofit. Her work weaves together inspirations from feminist studies and speculative fiction in building tangible hyper-local prototypes. She has also conducted work examining perspectives from post-communist Eastern Europe. More recently, she has been exploring how AI could be leveraged to help women and children seeking justice from gender based violence with South African non-profit Kwanele.

Ranjit Singh is a Researcher at the AI on the Ground Initiative of Data & Society Research Institute. His research interests lie at the intersection of data infrastructures, global development, and public policy. He examines the everyday experiences of people subject to data-driven practices and follows the mutual shaping

of their lives and their data records. More recently, he completed work that explores ways to actively think about data and artificial intelligence in regions outside of North America and Europe, while offering unique perspectives to investigate and understand the diverse ethics, politics, and everyday experiences of living with these technologies. He has published his research in venues such as the *Journal of South Asian Studies* and the ACM CHI Conference; he has presented his work at conferences including CSCW, 4S, AAA, and ECSAS.

Angelika Strohmayer is an Assistant Professor at Northumbria University's School of Design, and co-leader of the Design Feminisms Research Group. Her research interests lie at intersections of feminisms, technologies, and safety using trauma-informed, cocreative, and justice-oriented methodologies. One of her research strands relates to thinking about the role of technologies in relation to peoples post-digital safety. In this work, she aims to centre people and communities who are criminalised, stigmatised, or in other ways made marginal in society, exploring safety holistically, breaking down online/offline and digital/non-digital boundaries. Alongside this work, she is also co-editor of the 'Meaningful Design Processes' Forum in *Interactions Magazine* and published her first book 'Digitally Augmenting Traditional Craft Practices for Social Justice' in 2021. She regularly authors and co-authors publications in interdisciplinary venues in Human-Computer Interaction, Social Science, and Security venues.

Ethan Zuckerman is associate professor of public policy, information and communication at the University of Massachusetts at Amherst and director of the Initiative for Digital Public Infrastructure. His research focuses on the use of media as a tool for social change, the use of new media technologies by activists and alternative business and governance models for the internet. He is the author of *Mistrust: How Losing Trust in Institutions Provides Tools to Transform Them* (2021) and *Rewire: Digital Cosmopolitans in the Age of Connection* (2013). With Rebecca MacKinnon, Zuckerman co-founded the international blogging community Global Voices. It showcases news and opinions from citizen media in more than 150 nations and 30 languages, publishing editions in 20 languages. Previously, Zuckerman directed the Center for Civic Media at MIT and taught at the MIT Media Lab. In 2000, Zuckerman founded Geekcorps, a technology volunteer organization that sends IT specialists to work on projects in developing nations, with a focus on West Africa. Previously, he helped found Tripod.com, one of the web's first "personal publishing" sites.

4 PANEL FORMAT & AUDIENCE ENGAGEMENT

Interactivity and audience participation are major goals of this panel. Thus, we opt for a format that encourages audience engagement. The event consists of five panelists and three moderators (one leading the discussion and two actively monitoring the live chat). To engage with a broad audience, the panel will adopt an online format to allow both panelists and attendees the opportunity to join virtually and interactively engage with the panelists and attendees. The structure of the event would be as follows:

- **0-5 mins:** Allow participants to enter the online space and get settled

- **6-15 mins:** Introduction of the speculative design format (see box)
- **16-45 mins:** Moderated cycles of discussion between the panelists and audience
- **46-60 mins:** Self-reflection and audience-driven engagement with guiding questions about speculative futures
- **61-70 mins:** Summary of key insights from the discussion and conversations about next steps forward
- **71-75 mins:** Closing remarks and resource sharing to maintain post-event connections

To begin our moderated segment, panelists will offer brief introductions to familiarize the audience with their research and views on what they consider to be significant challenges and opportunities for accountability in algorithmic systems. Panelists will have the option of supporting their introductions with 1-2 slides. Discussions will be grounded in a speculative design lens to shape the narrative of future possibilities to something we have the ability to shape. Core questions would be rooted in the possibilities of "what if" to open up dialogue, debate, and discussion on current realities and alternative futures. Exemplar questions include: "What is the most desirable future? In the next 10 years? In the next 5 years? What can be achieved today for tomorrow? What would we *not* want to see from the future?". During the segment, there will be a live Q&A session where audience members will be encouraged to pose questions to the entire panel, thus leveraging the diversity in audience expertise who could further challenge and broaden perspectives. Audience members will also be encouraged to submit comments or questions throughout the entire event. Moderators will monitor questions or comments submitted using the chat feature to allow for equal and fair inclusion. Finally, time would be dedicated for reflection on critical questions for future directions where audience members would be asked to quietly self-reflect before delivering closing remarks and giving attendees the opportunity to connect. After the session, audience members would be able to continue discussions via Slack and build community with others who are committed to accountable AI systems.

"[Speculative design] thrives on imagination and aims to open up new perspectives on what are sometimes called wicked problems, to create spaces for discussion and debate about alternative ways of being, and to inspire and encourage people's imaginations to flow freely. Design speculations can act as a catalyst for collectively redefining our relationship to reality". [7]

5 PANEL OUTCOMES

At the conclusion of the panel, we endeavor to identify pressing issues that are important to broader research, development, and policy community members. We aim to first synthesize the insights and dialogues from the discussion in a blog post then subsequently develop a short position paper which would be submitted as an ACM magazine contribution (e.g. Interactions or Communications).

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